

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

Commented [U1]: Please enter the name in Arial 12 font

2. Subject Name: Spectroscopic Methods of Structure Determination

L-T-P: 3-1-0

Credits: 4

Commented [U2]: Please enter the details (use Arial 12 font)

3. Pre-requisites: Nil

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. Syllabus and reference books:

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Syllabus:

Infra-Red Spectroscopy: (4L)

Electromagnetic spectrum and applications of IR light; Instrumentation and various sample preparation techniques; Differences between UV and IR Spectrum; different parts of IR spectrum; Discussion on Hooke's Law and Dipole Moment; Functional group identification; Influence of electronic parameters on the peak position; Special focus on the hydrogen-bonding effects, Solvent effects.

Mass Spectrometry: (10L)

Different instrumentation (mass analyzers) and ionization techniques: EI (1 h), CI, FAB (1h), ESI (1 h), MALDI (1 h), etc. Exact mass, atomic weight, isotope distribution calculation, intensity calculation, exact mass calculation, DBE, nitrogen rule, etc.

Fragmentation: alkane, alkene, aryl alkane, ether, amine, alcohol, sulfide, McLafferty rearrangement, ketone, aldehyde, amino acids, etc.

UV Spectroscopy: (2L)

Nuclear Magnetic Resonance Spectroscopy: (31L)

Fundamentals of NMR: (10L)

Nucleus, how it behaves in B_0 , vector model, energy separation equations and precessional motion, NMR sensitivity and its comparison with IR and UV (Boltzmann distribution), net magnetization. Magnetogyric ratio, relative abundance and I value of important nuclei. Using these values, a comparative sensitivity of these nuclei towards NMR. Criteria for a nucleus to be NMR active.

What is pulse and physical meaning of 90° - and 180° pulses.

Relaxation (T_1 and T_2) physical interpretation: spin-lattice and spin-spin relaxation, Why T_1 relaxation time is more important than T_2 , how relaxation time effect the signal intensity of quaternary, tertiary, secondary and primary carbon atom, plot of [T_1 , T_2] vs correlation time, mechanism of spin-lattice relaxation.

NMR Scale and δ -value in ppm, Integration, reference solvent, CW and FT NMR, NMR signal recording as FID for FT NMR, how an NMR machine looks like.

Chemical Shift: (6L)

Dependence of σ -on several factors, chemical shift value in general for all class of compounds, chemical shift and EN value in a table, effect of EN and distance, Effect of ring current, Dependence of chemical shift on Concentration, temperature, solvents etc. Magnetic anisotropy (alkene, alkyne, carbonyl, benzene ring, EDG & EWG, heteroaromatic compounds, annulene, porphyrin, non-aromatic and antiaromatic systems), various substituents effect on alkene (α , β -unsaturated systems), anisotropy in cyclohexane and other cyclic systems, examples of several NMR spectra. Chemical shifts for other nuclei.

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Spin-Spin Coupling: (8L)

Direct vs indirect spin-spin coupling, *Indirect spin-spin-coupling*: orientation of electron through one and two bonds, meaning of nJ sign for nJ coupling, why it is called scalar coupling, selection rule. Total spin (m_I) for a group of nuclei, Satellite peak. Limits of simple selection rules, $\Delta v > J$ and $\Delta v < J$. Spin System and Magnetic Equivalence, AB, ABq, AMX, A₂B, A₂X, A₂B₂, AA'BB', AA'MM'XX', etc. Calculation of coupling constant from ABq and other 1st order multiplets, Rules to interpret the multiplet spectra, Vicinal Coupling, Factors influencing vicinal coupling: 1D-COSY, decoupling, understanding decoupling using vector model, homonuclear spin-spin decoupling, heteronuclear spin-spin decoupling, BB decoupling

¹³C NMR: (4L)

Introduction, comparison between proton and ¹³C NMR, sensitivity, proton decoupled ¹³C NMR, how relaxation time effect the signal intensity, DEPT

Structural elucidation: (3L)

Applications of analytical and spectral data generated from IR, Mass and NMR spectra in on structure determination.

Reference Books:

1. NMR Spectroscopy: Basic Principles, Concepts, and Applications in Chemistry by Harald Gunther; Third Edition; Wiley VCH 2013
2. Spectrometric Identification of Organic Compounds, Robert M. Silverstein, Francis X. Webster, David J. Kiemle, David L. Bryce; 8th Edition; Wiley 2014
3. Fundamentals of Molecular Spectroscopy By Colin N. Banwell, Fourth edition; McGraw-Hill 2017
4. Introduction to Spectroscopy by Donald L. Pavia, Gary M. Lampman, George S. Kriz, James R. Vyvyan; Fifth Edition; Cengage Learning 2015

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Infra-Red Spectroscopy	4 h
2.	Mass Spectrometry	10 h
3.	UV Spectroscopy	2 h
4.	Nuclear Magnetic Resonance Spectroscopy	31 h
5.	Tutorials	13 h
Total number of hours		60 h

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